



Koneru Lakshmaiah Education Foundation

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OPERATING SYSTEMS AND SYSTEMS PROGRAMMING (25CS2104E)

2026–27, Term-I

Project Problem Statement Submission Form

Section / Team Details

Section No.: 4

Team No.: 20

Project Title: Linux IPC Using PIPES and FIFOs

Team Members

Roll Number	Student Name	Signature
2520030248	Rishi Varma K	
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2520030420	Hitesh Reddy K	

1. Abstract

Argument Passing is a Linux-based system programming project that demonstrates how values are passed from the command line to a C program. The project uses the `argc` and `argv` parameters of the `main()` function to receive and process user-supplied arguments. It explains how the operating system provides command-line arguments to a newly executed process and how the program accesses them. The solution is implemented in C and executed in a Linux environment using standard compilation and execution commands. The expected outcome is a simple, working program that accepts different arguments, displays them, and demonstrates the basic relationship between process execution and command-line input.

2. Problem Statement

Programs often need input at the time of execution without requiring interactive input. This project addresses the problem of passing values to a program through the Linux command line. It demonstrates how command-line arguments are supplied when a process is started and accessed using `argc` and `argv`. The project is relevant to Operating Systems and Systems Programming because it connects process execution, program startup, and user input with C programming in a Linux environment.

3. Objectives

1. Demonstrate passing command-line arguments to a C program in Linux.

2. Use argc and argv to receive and access arguments.
3. Compile and execute the program using Linux system programming tools.
4. Understand the relationship between process execution and command-line input.

4. Proposed Methodology

The project is implemented in C on Linux/Ubuntu. The program defines main() with argc and argv parameters, reads the number of arguments, and accesses each argument using argv. The source code is compiled using GCC and executed from the terminal with different command-line inputs. The program validates the available arguments and displays them along with their positions, demonstrating the flow from command-line execution to program processing.

5. Operating Systems Concepts / Linux APIs Used

OS Concept / Linux API / System Call	Purpose in the Project
argc and argv	Receive and access command-line arguments.
Process Execution	Understand how arguments are supplied when a program starts.
main() Function	Entry point that receives command-line arguments.
Linux Terminal	Provide arguments and execute the program.
GCC Compiler	Compile the C source code into an executable.
POSIX/Linux Environment	Provide the platform for program execution and testing.

6. Individual Contribution

Roll Number	Student Name	Individual Responsibility
2520030435	Shaik Juneeth	Design and implementation of argument passing program.
2520030415	V Ashish Reddy	Linux execution, testing, and GitHub management.
2520030497	D Venkata Mahith	Documentation, output verification, and presentation.

7. Tools / Platforms / Software Used

Tool / Platform / Software	Purpose
Linux / Ubuntu	Operating system and terminal environment.
C / C++	Implementation of the argument passing program.
GCC	Compilation of the C source code.
VS Code	Source code editing and project development.

Git / GitHub	Version control and project repository management.
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8. Expected Outcome

The completed system will accept command-line arguments during program execution, identify the number of arguments, and display each argument with its position. It will successfully compile and run in Linux using GCC, providing a clear practical demonstration of argument passing, process execution, and basic system programming concepts.

9. GitHub Repository

GitHubRepositoryLink:

<https://github.com/kolanuvadarishivarma-cell/KLH-CSE-2026-2027-Team-20-Linus-IPC-Using-Pipes-FIFO>

Repository Created: Yes

☒ Yes ☐ No

Faculty Name and Signature

Faculty Name: _____

Signature: _____

Remarks: _____

Faculty Use

Project Approved: _____

☐ Yes ☐ No

Date: _____

Faculty Signature: _____